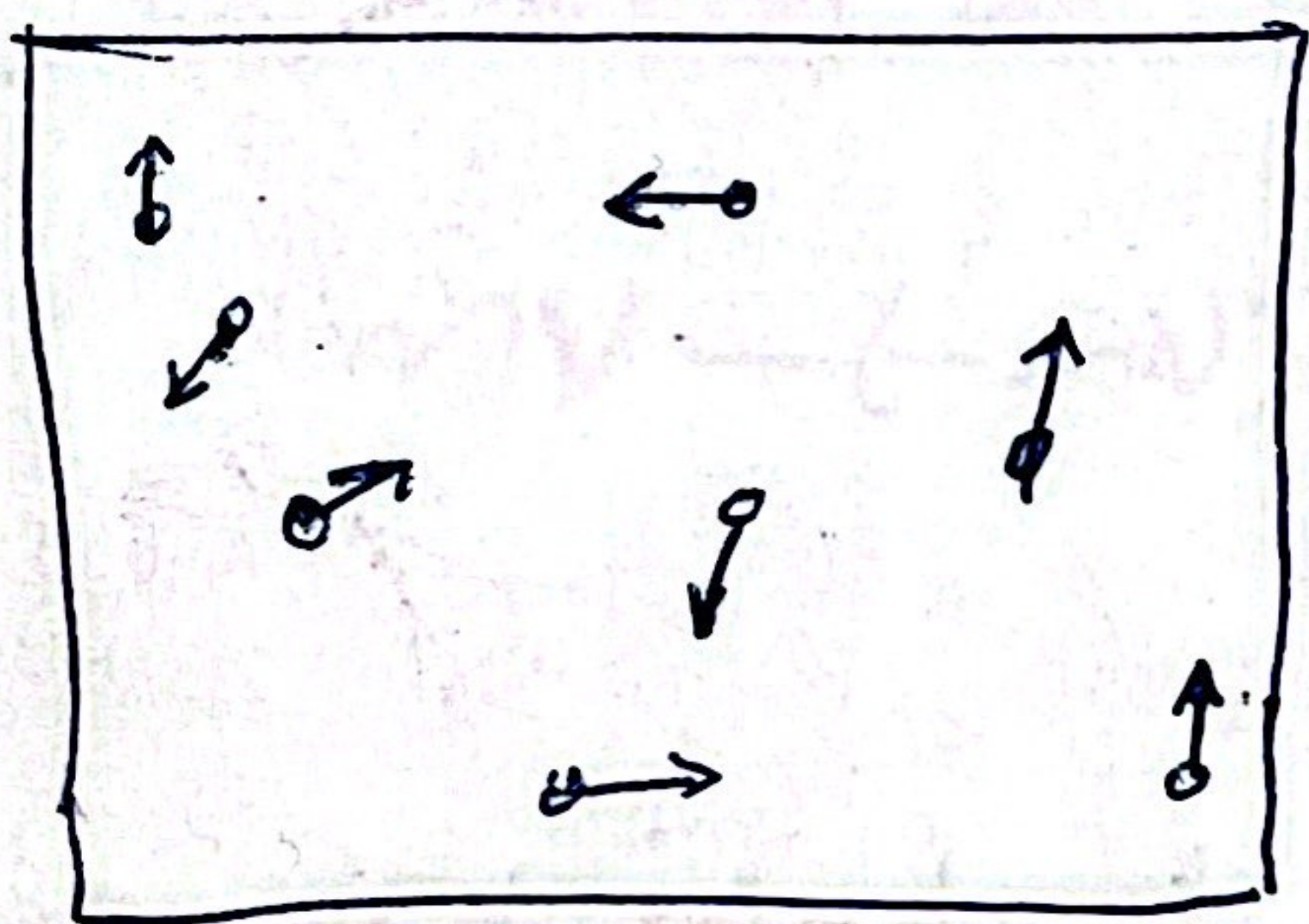


Kinetic Theory.

According to the Kinetic theory, molecules in a gas fly about in all directions, frequently colliding with each other and with their containing walls.



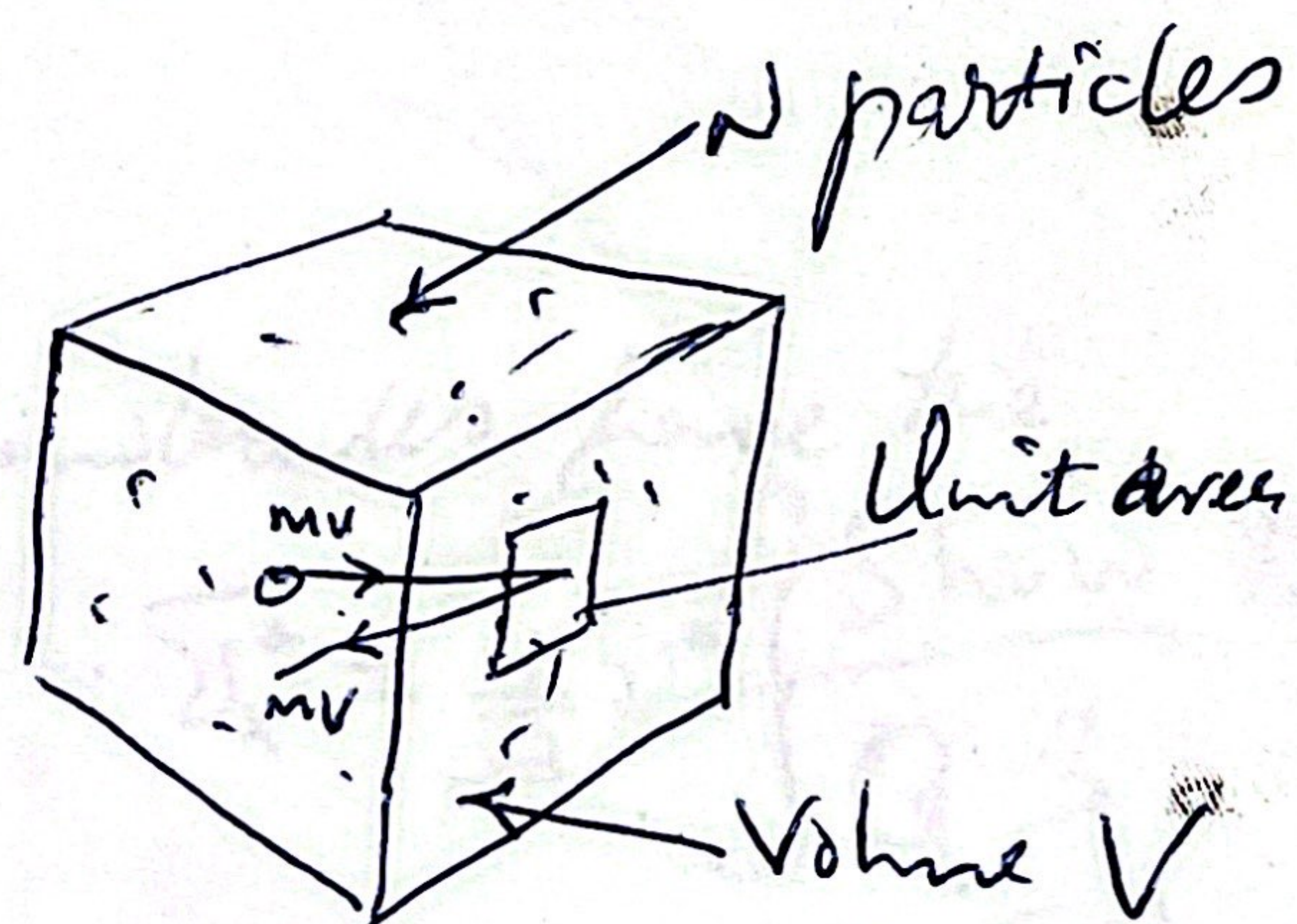
Assumptions:

- Each molecule spends most of its time in free motion between collisions and only a small fraction of its time in interaction with other molecules. This assumption is a good one for most actual gases at ordinary temperatures.
- Each collision between two molecules or between a molecule and a container wall is an elastic collision, one in which K.E is conserved.
- Newton's laws govern the motion and collision of molecules

We have already discussed the concept that heat energy is associated with molecular motion, and that the higher the temperature the greater the K.E. of the molecules.

Kinetic theory gives us a means of predicting the pressure on the wall of a container of gas. Every time a particle rebounds from the surface, it experiences a change of momentum, which by the laws of mechanics provides a force. The change in particle momentum is

$$mV - (-mV) = 2mV$$



To find pressure, it is necessary to add up the effects of all the particles with their various directions and speeds. Suppose N is the total number of molecules in a container of volume V . The number per unit volume is

$$n = \frac{N}{V}$$

The kinetic energy of each particle of common speed is

$$E_k = \frac{1}{2} mV^2 =$$

A collision occurs once every round trip of the molecule; hence the time interval between successive collisions is

$$\Delta t = \frac{2L}{V_x}$$

The average force (averaged over the time Δt) that is exerted on the right hand wall by collision of this single

molecule is

$$\bar{F} = \frac{\Delta P}{\Delta t} = \frac{2mV_x}{2L/V_x} = \frac{mV_x^2}{L}$$

Since N molecules contribute to the total force then

$$F = \frac{Nm\bar{V}_x^2}{L}$$

We have chosen $\bar{V}^2$ since not all the molecules have the same velocity. If $\bar{V}_x^2 = \bar{V}_y^2 = \bar{V}_z^2$ then $\bar{V}_x^2 = \frac{1}{3}\bar{V}^2$ } Show *
So that

$$F = \frac{Nm\bar{V}^2}{3L}$$

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{F}{L^2} = \frac{Nm\bar{V}^2}{3L^3}$$

Since $V = L^3$ volume, then $PV = \frac{1}{3} Nm\bar{V}^2$

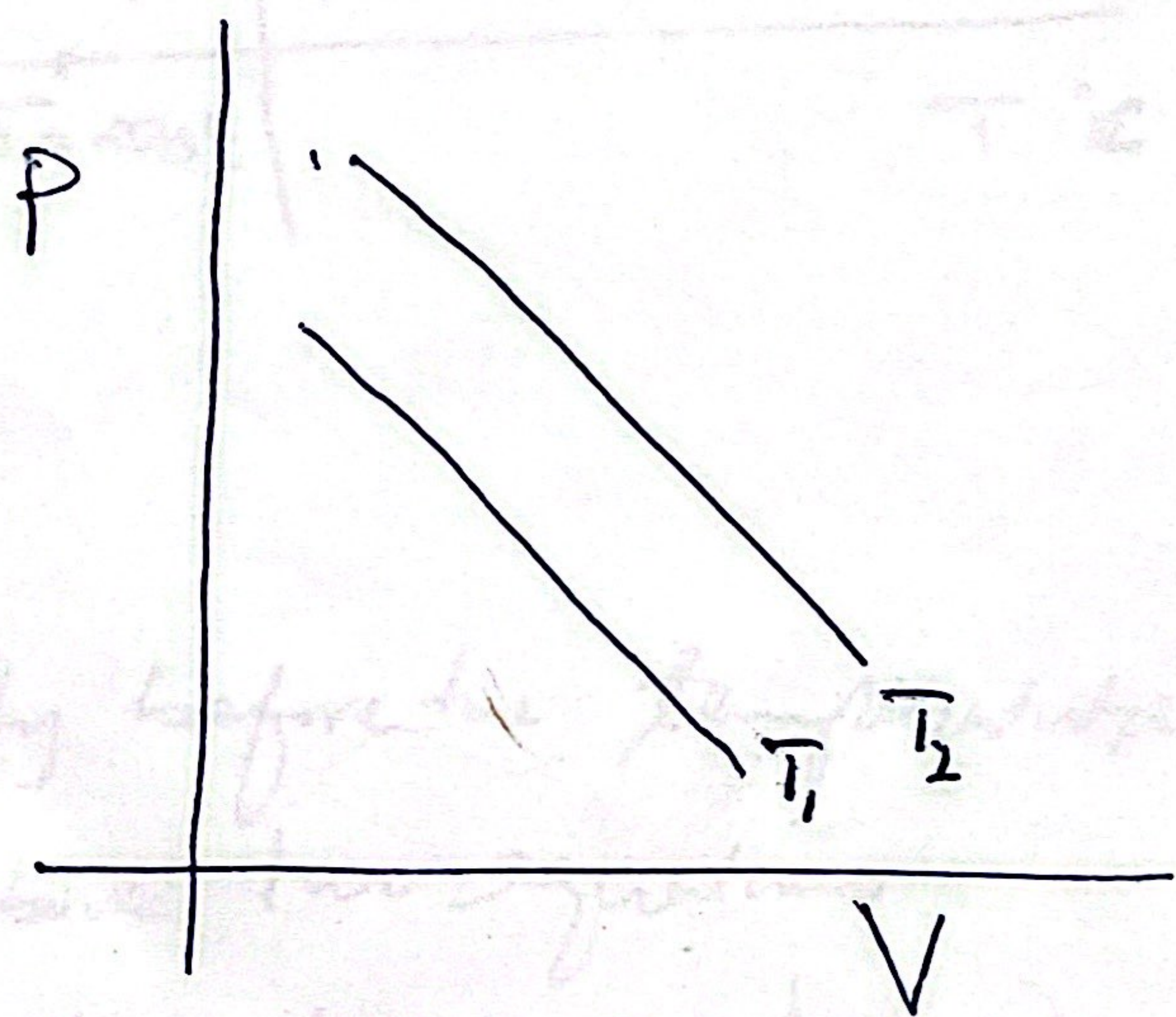
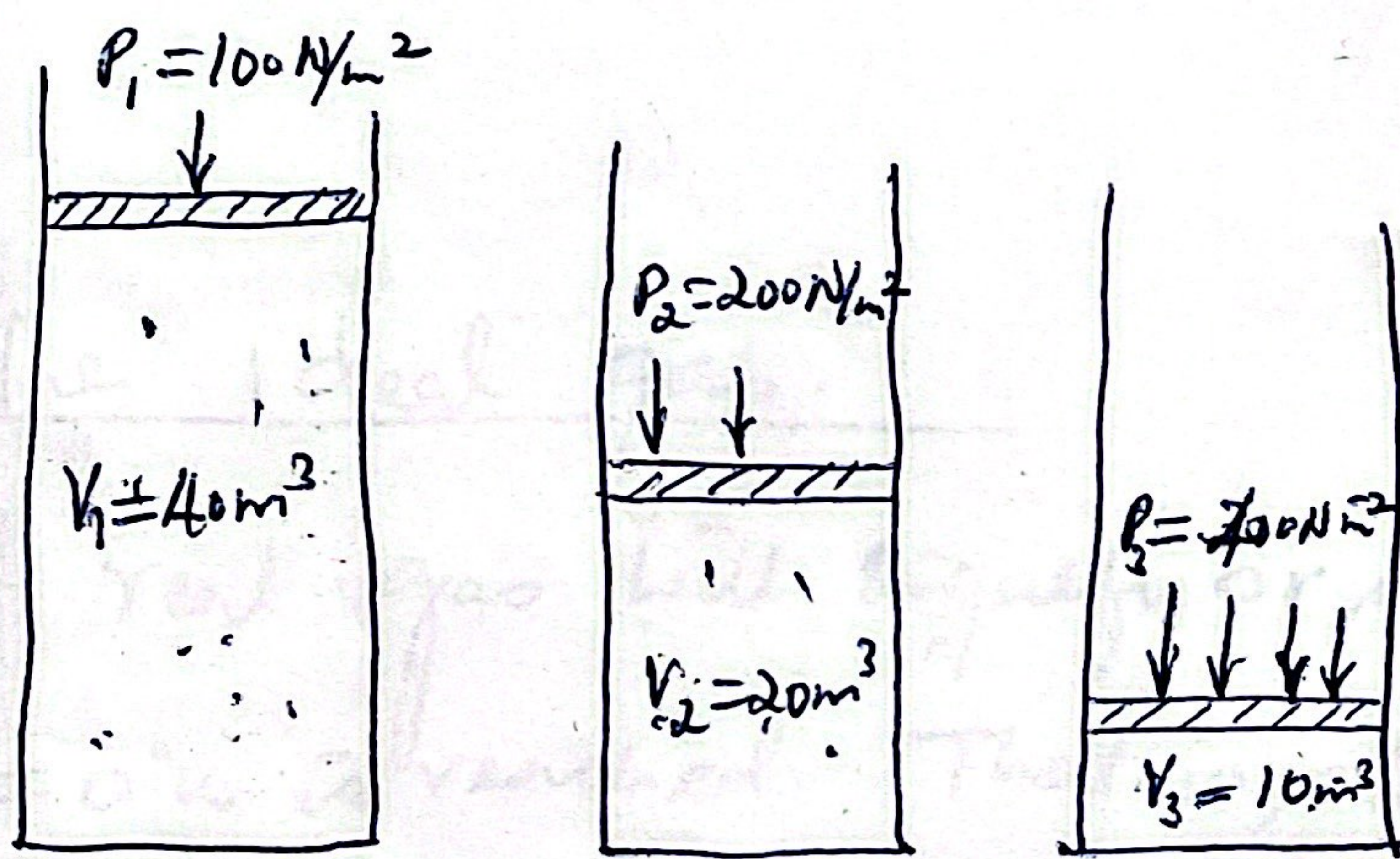
Inserting the expression for $\overline{K.E} = \frac{1}{2} m\bar{V}^2$ then

$$PV = \frac{2}{3} N \overline{K.E}$$

~~This is the ideal gas law.~~ It describes the microscopic properties of a gas.

Gas laws.

In 1662, Robert Boyle, an English Chemist discovered that the volume of a gas confined to a cylinder is inversely $\propto$ to the pressure exerted on the gas through piston as long as temperature remains constant.



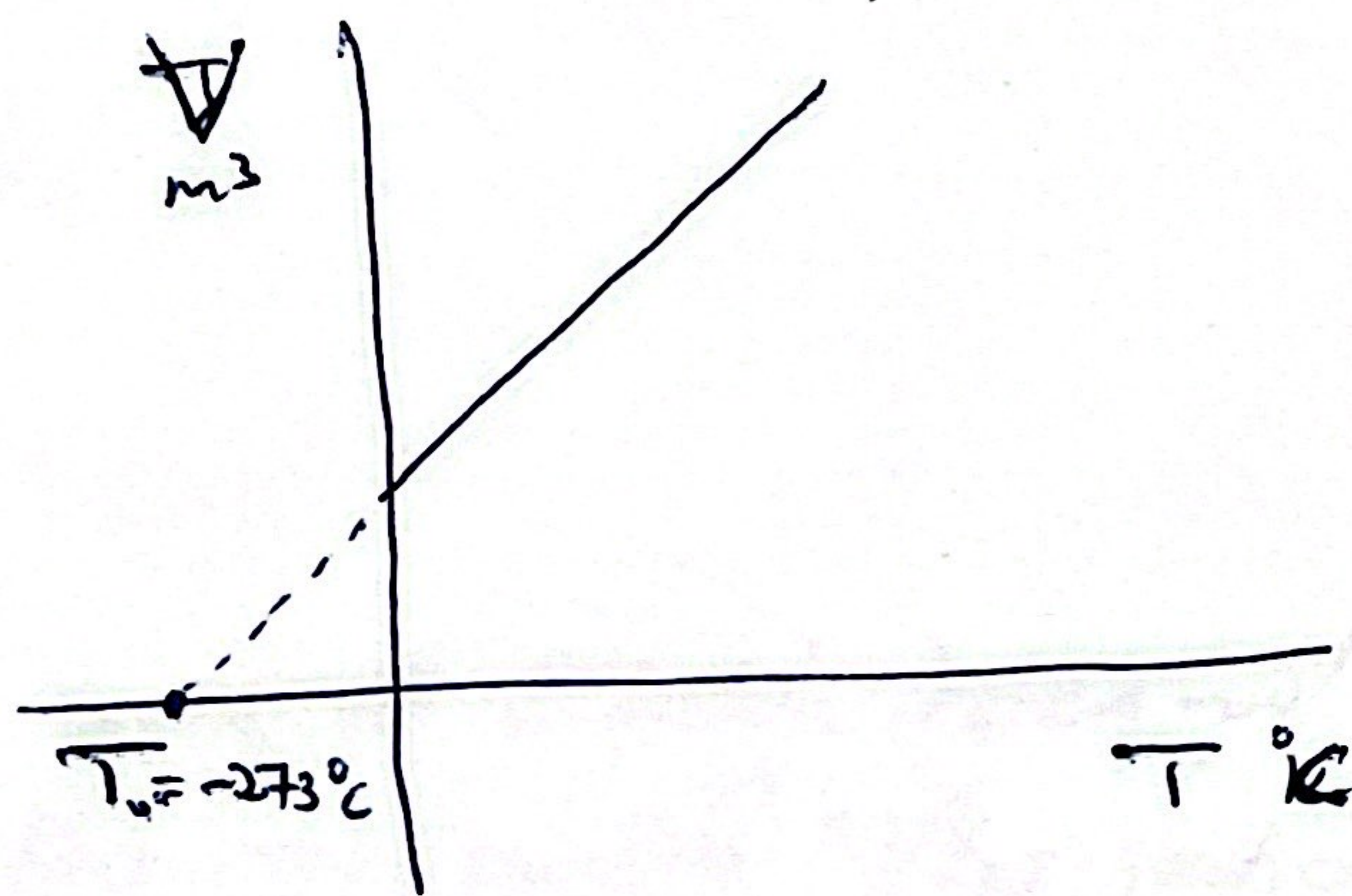
This rule is obeyed regardless of the shape of container, or the area of piston, or the nature of the gas. The term Boyle's law is given to this fact which can be expressed as

$$P \propto \frac{1}{V} \quad \text{or} \quad PV = \text{constant} \quad (T \text{ constant}).$$

The French physicists Jacques Charles and Joseph Louis Gay-Lussac independently discovered the relationship between the temperature of a gas and its volume (at constant pressure) and so extended Boyle's law. This relation

States that the "volume of a gas is proportional to its temperature," $V \propto T$ or

$$\frac{V}{T} = \text{constant}.$$



The Ideal gas.

A real gas will liquefy or solidify before the temperature $T = 0^\circ\text{K}$ is reached. That is, the above two equations are only approximate descriptions of real gases. An ideal gas obeys Boyle and Charles-Gay-Lussac law perfectly.

Combining the two equations

$$PV = \text{constant} \quad \text{and} \quad \frac{V}{T} = \text{constant}.$$

We can write the equation of state for an ideal gas

$$\frac{PV}{T} = \alpha \quad (\text{ideal gas}).$$

α is a constant and depends on the molecular properties of the gas. ~~For~~ Real gases, at low gas densities, are described by this equation.

The constant α is directly proportional to M , ^{mass of gas} and inversely proportional to m the molecular mass, or

$$\alpha \propto \frac{M}{m}$$

$$\Leftrightarrow \alpha = R \frac{M}{m}$$

R is a constant called Universal gas constant which does not depend on the specific properties of the gas used. The ratio of mass of a gas M to its molecular mass m is the number of moles n of the gas.

$$\text{number of moles} = n = \frac{M}{m}$$

Thus, the ideal gas law becomes

$$PV = nRT$$

The equation of state.
To be cont'd

Note The temperature must be given in $^{\circ}\text{Kelvin}$ ($^{\circ}\text{K}$).

The quantities P , V , T and n can be measured. R is explicitly determined to be

$$R = 8.314 \text{ J/mole } ^{\circ}\text{K}$$

The ideal gas law is very important in that it unites the mechanical concepts of pressure and volume with the concept of temperature.

In fact we have in this law our fundamental physical definition of temperature:

$$T = \frac{PV}{nR}$$

We can use this relation to determine exactly the temperature of a medium by constructing a gas thermometer. Such an instrument consists of a definite volume of gas and a method for measuring pressure. Both the volume and the pressure can be measured and temperature therefore determined. A gas thermometer is useful over a wider range of temperature than a mercury thermometer because mercury freezes at -38.9°C and boils at 356.6°C . In principle at least, a helium thermometer could be used down to the temperature at which it liquefies, 4.2°K .

#

We can unify the ideal gas law $PV = nRT$ and the microscopic properties of a gas law $PV = \frac{2}{3} N \overline{KE}$ to obtain

$$\frac{2}{3} N \overline{KE} = nRT$$

$$\text{or } \overline{KE} = \frac{3}{2} \frac{nRT}{N}$$

The ratio $\frac{N}{n} = n_2$ of molecules per mole = Avogadro's number = 6.02×10^{23}

$$\text{Let } k = \frac{nR}{N} = \frac{8.314 \text{ J/mole}\cdot\text{K}}{6.02 \times 10^{23} \text{ molecules/mole}} = 8.62 \times 10^{-5} \text{ eV/}^\circ\text{K} \\ (1.38 \times 10^{-23} \text{ J/K})$$

1k k = physical constant called the Boltzmann constant.

$$\overline{KE} = \frac{3}{2} kT$$

This equation gives the connection between the macroscopic temperature and the microscopic kinetic energy

Recall that $\overline{KE} = \frac{1}{2} m \overline{v}^2$

Therefore $\frac{1}{2} m \overline{v}^2 = \frac{3}{2} kT$

$$\Rightarrow \overline{v}^2 = \sqrt{\frac{3kT}{m}}$$

This gives the root-mean-square (rms) velocity of molecules

$$v_{rms} = \sqrt{\frac{3kT}{m}}$$

Seen earlier in diffusion.

The velocities of the molecules are actually distributed according to a function called the Maxwellian distribution

Example

Compute rms velocity for an oxygen molecule at room temperature.

Sol.

Since room temperature is approximately 20°C , we use the absolute temperature $T = 273 + 20 = 293^{\circ}\text{K}$.

The mass of an oxygen molecule approximately 32 Amu ;

thus, $m = 32 \times 1.66 \times 10^{-27} \text{ kg} = 5.3 \times 10^{-26} \text{ kg}$. Therefore

$$V_{\text{rms}} = \sqrt{\frac{3KT}{m}} = \sqrt{\frac{3 \times (1.38 \times 10^{-23} \text{ J/K}) \times 293^{\circ}\text{K}}{5.3 \times 10^{-26} \text{ kg}}} = 478 \text{ m/s.}$$

Which is very fast. Of course, molecules do not travel very far at these speeds because of frequent collisions that change the directions of motion. Therefore it takes a certain amount of time for the scent of perfume from a bottle uncorked in one part of the room to penetrate (or diffuse) to the other side of the room.

Exercise.

Compute rms velocity for an ~~oxygen~~ nitrogen molecule at room temperature [Ans. 510 m/s]

Hint: $M = 28 \times 1.66 \times 10^{-27} \text{ kg}$; $K = 1.38 \times 10^{-23} \text{ J/K}$; $T = 293^{\circ}\text{K}$.

Avogadro hypothesis.

A mole of a substance (element or compound) is defined to be an amount that has a mass in grams equal to the molecular mass of the substance expressed in A.M.U. Thus, a mole of hydrogen has a mass of 2g, a mole of Oxygen has 32g, water, 18g.

A mole of any substance contains exactly the same number of molecules as a mole of any other substance. This number is called the Avogadro's number:

Avogadro's number $\equiv N_0$ = no. of molecules

$\equiv 6.022 \times 10^{23}$ molecules/mole.

The number of molecules N in a sample of mass M of a substance that has a molecular mass m is given by

$$N = \frac{M}{m} N_0$$

Where M/m is the fraction of a mole present in the sample.